

Hai Bao

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Professional Experience

Sr. Data Scientist | Morningstar Sustainalytics | Jan 2023 - Mar 2026

- Streamlined data ingestion workflows to process 400,000+ news articles daily by engineering an automated ETL process using Python, which included data validation, reducing manual data handling and improving analysis turnaround.
- Optimized business support operations by integrating a GPT-powered LLM solution (OpenAI API), successfully reducing daily manual research tasks by 25% (1,000 articles) for the ESG Research team.
- Enhanced management's decision-making capabilities by developing a dashboard in Power BI to track ESG Research workload distribution, enabling data-driven insights into team productivity and bottleneck identification.
- Translated complex AI model performance metrics (Accuracy, Precision, Recall) for non-technical Senior Leadership, aligning cross-functional understanding.
- Led a team of 4 Data Scientists as part of the 'Digital Innovation' team for 6 months before a re-org moved the team onto the Data Science/AI team.

Data Scientist | Morningstar Sustainalytics | Jun 2022 - Jan 2023

- Optimized the analysis of public Sustainability/Annual Reports by developing an EU Taxonomy Detection tool, reducing manual review time by 25% through the implementation of a RAG-based data model to automate disclosure detection.
- Enhanced business analysis accuracy by reducing noise by 50% for the Thematic Crawler by engineering a multi-layer text mining framework to streamline data processing.
- Engineered rigorous data validation and cleaning pipelines for stakeholder-provided inputs, eliminating data discrepancies and preventing downstream reporting errors.

Business Analyst | Morningstar Sustainalytics | Nov 2021 - Jun 2022

- Orchestrated the deployment of customized Thematic Crawler applications using Python & SQL, resulting in a 75% reduction in manual data extraction time, by facilitating cross-functional requirement-gathering sessions to align technical capabilities with diverse business needs.
- Delivered actionable insights via Excel reporting to support strategic decision-making for 5 business units
- Engineered a monthly PowerPoint analytical framework for SLT to track FTE savings against Bing API spend, successfully preventing budget overruns through proactive cost monitoring.

Technical Skills

- **Languages:** Python, SQL, VBA
- **Data Science and ML:** scikit-learn, XGBoost, K-means Clustering, Regression Forecasting, Text Mining, Data Validation
- **Generative AI and LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), OpenAI API, Prompt Engineering
- **Data Engineering and Databases:** ETL Pipelines, PostgreSQL, Data Ingestion, Data Cleaning
- **Data Visualization and Analytics:** Plotly, Power BI, Advanced Excel, Dashboards
- **Tools and Infrastructure:** AWS, Docker, Git, GitHub, MLB StatsAPI

Projects

Adventure Cycle Sales Dashboard

- Designed and deployed an interactive Power BI dashboard analyzing complex retail sales datasets to track critical financial and operational KPIs.
- Engineered data models and custom DAX metrics to deliver dynamic visualizations, enabling stakeholders to identify high-performing product categories and seasonal sales trends.

Walmart Customer Segmentation

- Applied unsupervised machine learning by executing the K-means Clustering algorithm on large-scale Walmart retail transactions to isolate distinct consumer behavioral archetypes.

Rogers Centre Attendance Forecast

- Built an end-to-end data pipeline to ingest, clean, and load historical baseball attendance records from the MLB StatsAPI into a local PostgreSQL database.
- Developed a predictive regression model using XGBoost to forecast stadium attendance, optimizing accuracy by integrating external feature variables such as weather forecasts, team standings, and promotional schedules.

Education

- **B.S. in Statistics**, University of Toronto
- **Databricks Fundamentals**, Databricks